

HYPERSENSITIV IMAGING FOR INTRAOPERATIVE BRAIN TUMOR DETECTION: FROM BENCHMARKS TO PREDICTION

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ABSTRACT: Accurate delineation of tumor margins during neurosurgery is essential for patient survival. Current intraoperative guidance tools, such as pre-operative MRI-based neuronavigation or fluorescence-guided surgery (5-ALA), face limitations due to *brain shift* phenomena and subjective interpretation. In this scenario, *Hyperspectral Imaging* (HSI) emerges as an alternative paradigm: a real-time technology capable of discriminating tissues by analyzing their spectral signature far beyond the visible spectrum.

This work presents an in-depth study on the use of *Machine Learning* (ML) and *Deep Learning* (DL) algorithms for tumor segmentation in HSI images. The research begins with the replication and validation of the "InVivoBench" benchmark (based on the HELICoID project), implementing and comparing classifiers such as *Support Vector Machines* (SVM) (Cristianini and Shawe-Taylor, 2000), *Random Forests* (RF), and 1D *Deep Neural Networks* (1D-DNN) (Goodfellow et al., 2016) on *in-vivo* hyperspectral images. Starting from this *baseline*, the study analyzes the limitations of current approaches, specifically addressing the *label noise* problem that intrinsically affects the histological *ground truth*.

The work introduces and evaluates advanced aggregation strategies (*Ensemble Learning*), implementing *Soft Voting* mechanisms and a novel *Clinical Stacking* approach. Empirical results demonstrate that model aggregation mitigates variance, outperforming individual classifiers in terms of robustness. Central to this context is the analysis of Epistemic Uncertainty. Through the use of *Risk-Rejection* curves, we demonstrate that filtering high-entropy voxels allows for modulating the system's sensitivity. The outcome is a *risk-aware* approach: a decision support system for surgeons that not only classifies but also quantifies prediction reliability, paving the way for transparent and computationally robust assisted surgery.

DECLARATION OF ORIGINALITY: This paper constitutes an independent and critical re-evaluation of the InVivoBench benchmark. The original contribution lies in the design of a *Clinical Stacking Ensemble* architecture, specifically optimized to maximize oncological sensitivity. In parallel, a validation methodology based on epistemic uncertainty (*Risk-Rejection Curves*) is formalized, developed to intercept and reduce classification errors in *safety-critical* clinical contexts. The code for replicating the results of (Fabelo et al., 2019), as well as testing novel experiments, is available in the official GitHub repository of the project (Botta, 2026).

1 INTRODUCTION

1.1 Clinical Context and Problem Definition

In the field of oncological pathologies of the nervous system, gliomas constitute the most widespread class of primary tumors. Glioblastoma (GBM), in particular, represents the most critical variant due to its rapid evolution and tendency to expand beyond the visible boundaries of the main mass. The current intervention protocol aims for maximal tumor volume removal: data confirms that the extent of the resected mass is the factor that most significantly impacts the patient's lifespan and the stability of their condition.

The main problem for surgeons concerns the uncertainty in tissue classification at the lesion margins. In this transition zone, tumor cells mingle with healthy ones such that, under normal optical lighting, the two tissues are visually indistinguishable. This creates a need for balance: excessive removal can cause permanent damage to motor or cognitive functions, while insufficient removal leaves cellular residues that inevitably lead to rapid recurrence.

From a technological perspective, current support tools show structural vulnerabilities. Neuronavigation relies on Magnetic Resonance Imaging (MRI) acquired before the operation. However, once the surgical procedure begins, the brain undergoes physical deformation due to gravity and changes in internal pressure. This *brain shift* phenomenon introduces a registration error between the pre-operative dataset and the real tissue position (Fabelo et al., 2019), rendering digital maps geometrically inaccurate. Other solutions, such as ultrasound, often present a signal-to-noise ratio that is too low, while fluorescence-based techniques do not guarantee reliable detection in all operative scenarios.

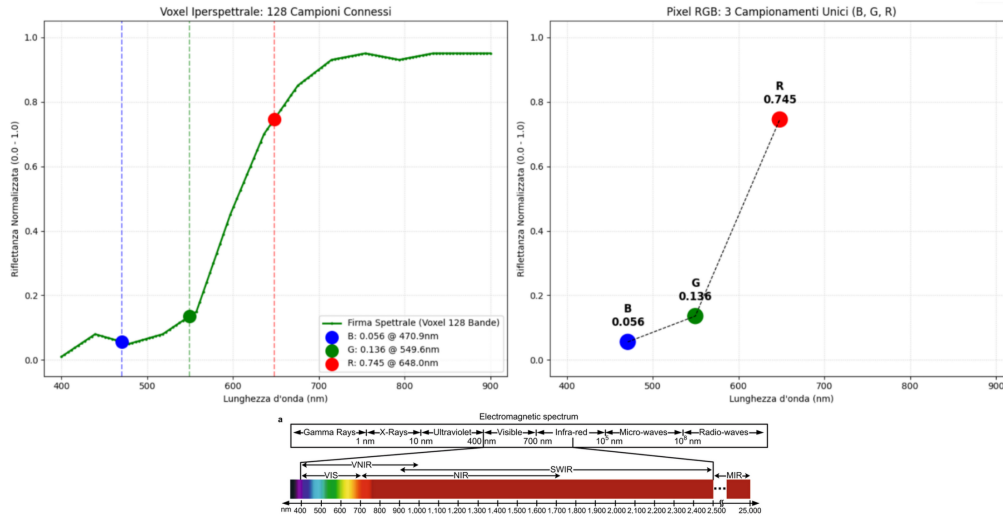


Figure 1: Comparison between a standard RGB pixel structure (discrete sampling) and a hyperspectral voxel (continuous sampling with 128 bands) in the VNIR range.

1.2 Hyperspectral Imaging (HSI) as a Solution

Hyperspectral Imaging (HSI) presents itself as an evolution of computer vision systems, capable of overcoming the information compression typical of RGB sensors. As illustrated in Figure 1, while a traditional pixel samples the light signal into only three discrete components, an HSI system acquires data through hundreds of narrow, contiguous spectral bands. In this scenario, the minimum unit of information is no longer a simple pixel, but a **hyperspectral voxel**: a volumetric element that associates a high-dimensional vector in the wavelength domain (λ) with each spatial coordinate (x, y).

This data architecture generates the so-called "hypercube", a 3D structure where the third dimension represents the tissue's spectral signature. From a computational classification standpoint, this signature is a biochemical descriptor of the imaged material. In intraoperative monitoring, the hyperspectral voxel is sensitive to variations in oxygenation levels and hemoglobin concentration. Since tumor metabolism drastically alters the spectral response compared to healthy parenchyma, it is possible to map these signal divergences to perform an "optical biopsy." This allows for automated tissue segmentation based purely on reflected data analysis, eliminating the need for physical sampling or direct contact.

1.3 State of the Art and Benchmarks

The evolution of HSI systems applied to neurosurgery found a fundamental catalyst in European initiatives such as the HELICoiD project. In this regard, the contribution of León et al. (2023) defined a rigorous testing infrastructure through the "InVivoBench" benchmark, designed specifically for the computational validation of *Machine Learning* models on *in-vivo* spectral signals. This benchmark systematically addresses issues related to inter-patient variability and signal consistency post-calibration, offering a standardized environment for the comparative evaluation of architectures such as *Support Vector Machines* (SVM), *Random Forest* (RF), and *Deep Neural Networks* (DNN).

Recently, a noteworthy benchmark that could redefine the state of the art in this field was introduced: the SLIMBRaIN benchmark by Martín-Pérez et al. (2025). The technical goal here is the integration of heterogeneous data streams: the combination of high-resolution RGB images with the informational density of spectral data. This *data fusion* approach aims to compensate for the sometimes limited spatial resolution of hyperspectral sensors, ensuring tissue segmentation that is spatially coherent and spectrally accurate.

SLIMBRaIN will not be the subject of this study; however, its potential for future research is highlighted.

1.4 Thesis Contributions and Structure

The work is structured as follows:

- **Baseline Replication:** Initially, the classification pipeline described in the InVivoBench benchmark (León et al., 2023) is implemented and validated. A standardized pre-processing chain is used, and the performance

of classical ML (SVM, RF, KNN) and DL (1D-DNN) models is evaluated on the classification of Tumor Tissue (TT), Healthy Tissue (Normal), Blood Vessels (BV), and Background (BG).

- **Critical Analysis:** The limitations of the baseline approach are analyzed, focusing particularly on the *label noise* problem, where the histological *ground truth* might not align perfectly with the spectral reality at the tumor margins.
- **Proposed Methodologies:** The integration of *Ensemble* models, combined with uncertainty quantifiers, is explored to improve prediction accuracy, particularly the detection sensitivity of the tumor class (TT).

2 MATERIALS AND METHODS

This chapter describes the data sources, the architecture of the sensors used, and the computational pipeline necessary to transform raw signals into valid inputs for the classification models.

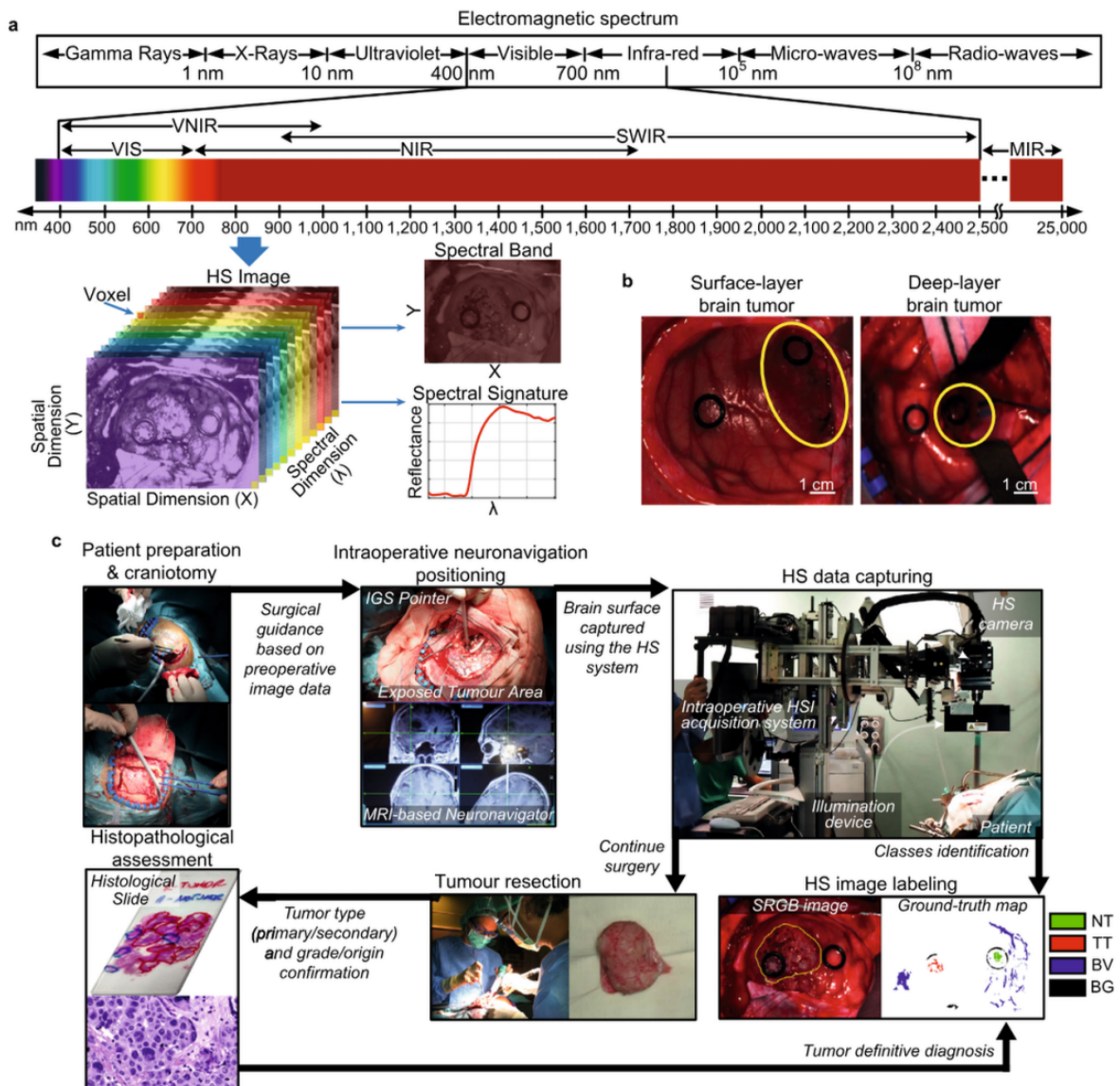


Figure 2: Acquisition flow diagram: from scene illumination to hypercube generation and spectral signature extraction for each spatial coordinate (adapted from León et al. (2023)).

2.1 Acquisition Systems

The reference dataset for this study is *InVivoBench*, derived from the activities of the European HELICoiD project (León et al., 2023). This database represents the current standard for hyperspectral neurosurgery, providing labeled *in-vivo* acquisitions. The generation of such data is based on two distinct technological paradigms:

- **Push-broom Systems:** Operate via linear scanning, ensuring very high spectral resolution (over 800 bands). Although they are the main contributors to *InVivoBench* due to their precision, the need for relative movement between the sensor and the scene introduces latencies and potential kinematic artifacts.
- **Snapshot Systems:** Designed for *real-time* integration, they capture the entire informational volume in a single exposure. This operational speed involves a trade-off, limiting the spectral resolution to about 16-25 bands.

2.2 From Spatial Geometry to the Spectral Domain

The acquisition process, summarized in Figure 2, transforms the optical response of the tissue into a complex data structure. Unlike traditional computer vision systems (RGB), where each (x, y) coordinate returns a three-channel discrete pixel, HSI generates a **Hyperscube** $I(x, y, \lambda)$.

2.2.1 The Information Unit: From Pixel to Voxel

The minimum element of this dataset is not a pixel, but a **hyperspectral voxel**. As highlighted in the comparison between RGB and HSI sampling, the voxel represents an informational volume that extends spatial resolution into the depths of the spectral domain. While an RGB pixel is a triad of isolated values, the voxel contains a continuous spectral signature: a high-dimensional vector describing the interaction of light with matter for every point in the scene.

2.2.2 Spectral Signature and Biochemical Discrimination

From a data analysis perspective, the spectral signature extracted from the voxel acts as a unique biochemical descriptor. In the VNIR range (400-1000 nm), the shape of the curve reflects critical physiological variables, such as oxygen saturation and hemoglobin density. Since tumor tissues (TT) exhibit altered metabolism and chaotic vascularization compared to healthy tissue (NT), their spectral responses show statistically significant divergences. The task of *Machine Learning* models is precisely to map these signal discrepancies to automate tumor margin segmentation.

2.3 Preprocessing and Data Conditioning Pipeline

Before feeding the voxels into the classifiers, a signal refinement process is necessary to eliminate instrumental biases. The implemented pipeline follows a deterministic four-stage approach:

1. **Radiometric Calibration:** To attenuate interference from the surgical lamp, each image *RI* (*Raw Image*) is expressed through reflectance $R \in [0, 1]$ using the relationship:

$$R = \frac{RI - DI}{WI - DI} \quad (1)$$

where *WI* (*White Image*) and *DI* (*Dark Image*) are the white and dark references, respectively.

2. **Smoothing and Spectral Filtering:** To mitigate high-frequency oscillations (*spikes*), a 1D convolution is applied at the voxel level. The use of a 5×1 convolutional kernel allows for regularizing the curve while maintaining the position of the absorption peaks.
3. **Pruning and Band Downsampling:** Due to the low efficiency of the hyperspectral sensor at the ends of the range, the first 56 and the last 126 bands are removed. The resulting hyperscube at $\lambda = 645$ is further sampled down to 128 channels to optimize model convergence without loss of generalization.
4. **Min-Max Normalization:** Finally, each spectral vector is normalized in the $[0, 1]$ interval. This *feature scaling* step ensures that illumination variance due to irregular brain morphology does not negatively influence classifier performance.

2.4 Critical Analysis of the InVivoBench Dataset

In this study, an in-depth quantitative and qualitative analysis of the *InVivoBench* dataset was conducted to understand the intrinsic characteristics of the data and the potential limitations influencing the performance of *Machine Learning* models.

2.5 Acquisition Protocol and Ethical Considerations

The collection of hyperspectral images follows a rigorous experimental protocol, structured to ensure maximum consistency between clinical evidence and the digital signal. The operative phases begin with surgical preparation (craniotomy and durotomy), necessary for direct exposure of the cerebral parenchyma. Once the field is stabilized, the HSI system is positioned to capture data in the area of interest. In this phase, the use of physical markers — specifically rubber rings — is fundamental for the spatial identification of *Regions of Interest* (ROI). The cycle concludes with tissue resection and subsequent histopathological analysis; the latter provides the definitive diagnosis that serves as the **Ground Truth** for training the classifiers.

However, the methodological integrity of the study must face a specific threat to external validity (**Threat to Validity 1 - TTV1**). The requirement of informed consent, while being an essential ethical pillar, acts as a selective filter on cases. This constraint could introduce a non-negligible *sampling bias*: the systematic exclusion of pediatric patients or subjects in advanced tumor stages, whose cognitive abilities are compromised, limits the representativeness of the dataset. From the perspective of model generalization, this imbalance implies that classifier accuracy might not reflect the clinical heterogeneity of the entire population affected by gliomas.

2.5.1 Labeling Procedure and Ground Truth Definition

Data labeling was performed using a semi-automatic approach supervised by expert neurosurgeons. The process uses the *Spectral Angle Mapper* (SAM) algorithm to select voxels spectrally similar to a reference voxel chosen by the physician, within a certain tolerance threshold.

This approach presents two significant methodological criticalities:

1. **Absence of sharp margins:** As shown in Figure 3, the *Ground Truth* (GT) maps are sparse and do not define continuous tumor boundaries. This makes the benchmark unsuitable for evaluating the completeness of surgical resection, limiting the analysis to *pixel-wise* classification rather than full semantic segmentation.
2. **Augmentation Bias (TTV2):** The use of the SAM algorithm to expand the label set tends to select voxels with high spectral similarity, acting as a filter that could exclude "anomalous" or heterogeneous tumor spectral signatures. This reduces intra-class variance in the training set, artificially simplifying the classification problem compared to clinical reality.

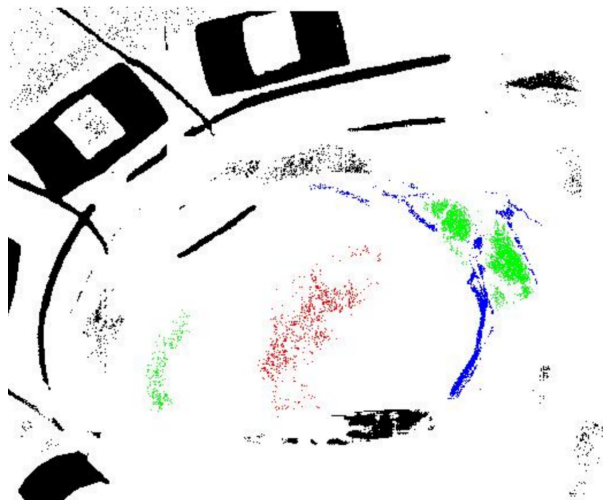


Figure 3: Example of a Ground Truth map. Note the sparse nature of the annotations and the absence of defined sharp resection margins for tumor tissue (TT).

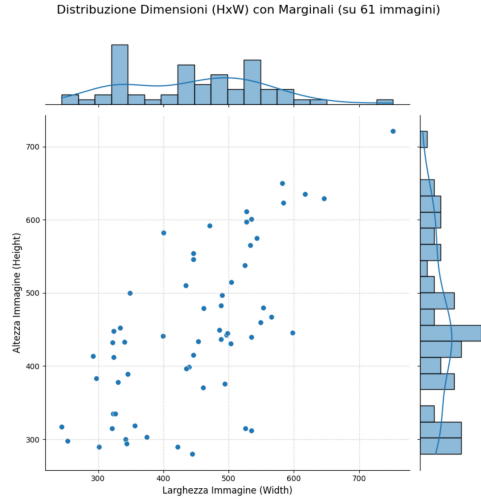


Figure 4: Distribution of spatial dimensions (Height vs Width) of the 61 hyperspectral images in the dataset.

2.5.2 Exploratory Analysis and Statistical Characterization of Data

The analyzed informational infrastructure consists of 61 hyperspectral acquisitions deriving from multiple surgical sessions. The preliminary investigation into the geometric properties of the data, visualized in the graph of Figure 4, reveals high heterogeneity in spatial resolution. The dimensions of individual hypercubes oscillate in a range between approximately 300×150 and 750×750 voxels; this notable variance is justified by the desire to maintain a high qualitative standard for the dataset, keeping only the regions of greatest interest.

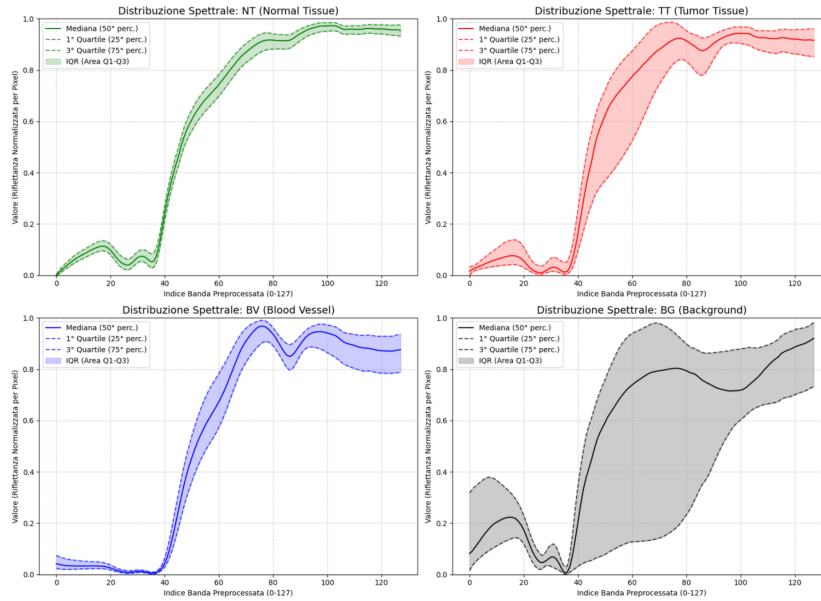
Beyond geometric variability, the analysis highlights a structural criticality regarding the distribution of labels, detailed in Table 1. The dataset exhibits an extremely marked *Class Imbalance* phenomenon. First, a massive presence of unlabeled data is observed: 92.89% of the total voxel volume (over 11 million units) lacks a defined *Ground Truth*. Among the labeled components, the *Background* (BG) class constitutes the predominant cluster, followed in descending order by healthy tissues (NT) and blood vessels (BV).

The most relevant problem for predictive models concerns the *Tumor Tissue* (TT) class. Despite its clinical centrality, tumor tissue represents only 0.29% of the overall dataset. The scarcity of data is further confirmed by the distribution across images: 28 out of 61 acquisitions are completely devoid of voxels belonging to the TT class. This asymmetry requires the adoption of specific sampling strategies or weighted cost functions to prevent the model from converging toward predictions biased toward overrepresented classes.

Table 1: Global voxel distribution per class in the dataset (12,709,334 total voxels).

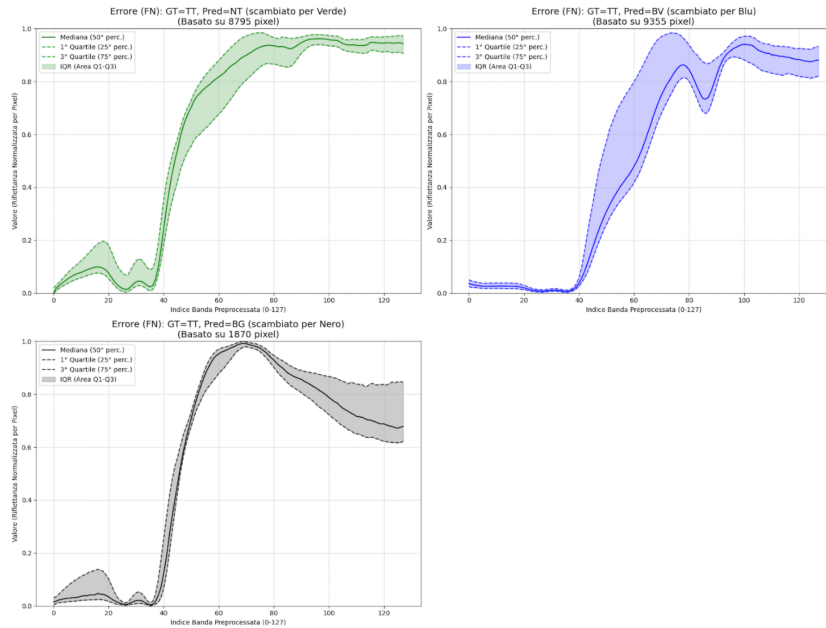
Label ID	Class	Voxel Count	Percentage (%)
0	Unlabeled	11,805,660	92.89
4	BG (Background)	445,540	3.51
1	NT (Normal Tissue)	308,328	2.43
3	BV (Blood Vessel)	112,407	0.88
2	TT (Tumor Tissue)	37,399	0.29

Analisi Statistica delle Firme Spettrali (Mediana e IQR)



(a) Statistical characterization (Median and IQR) of spectral signatures.

Analisi Spettrale Errori Falsi Negativi (FN) per SVM-L



(b) Spectral analysis of False Negatives (FN) generated by SVM-L.

Figure 5: Comparison between global signatures and error patterns: evidence of spectral ambiguity.

2.5.3 Spectral Characterization and Ambiguity Analysis

The investigation into spectral density and relative interquartile range (IQR), summarized in Figure 5a, allows for mapping the biochemical boundaries of the observed tissues. Although recurring morphological patterns are found — mainly related to hemoglobin absorption minima — pure class separability in the feature space remains a complex goal.

To investigate this criticality, a preliminary diagnostic analysis was conducted using a Support Vector Machine (**Linear SVM**). The objective was not absolute performance, but rather the identification of systematic model failures. Empirical evidence, illustrated in Figure 5b, revealed an unexpected datum: tumor voxels (TT) erroneously

classified as Blood Vessels (BV) or Normal Tissue (NT) do not show a degraded or uncertain signal. On the contrary, they exhibit spectral profiles that faithfully mimic the signature of the predicted class.

From a *Pattern Recognition* perspective (Bishop, 2006), this phenomenon indicates an **intrinsic spectral ambiguity**. This is not simply an inability of the classifier to generalize, but rather a stochastic overlap of signatures. For example, a hypervascularized tumor produces a reflectance response that perfectly mimics the spectrum of a blood vessel, making the two classes indistinguishable on a purely point-based basis.

3 Classifier Replication and Comparison

3.1 Validation and Data Leakage Prevention

The evaluation of classifier performance follows a rigorous validation protocol designed to test the models' generalization ability on subjects never seen during training. To this end, a **custom patient-level 5-fold Cross-Validation** procedure was implemented.

Unlike classical k -fold, which performs an arbitrary partition of the global dataset, the implemented logic ensures data isolation based on unique subject identifiers (*Patient IDs*). For each fold, the patient pool is divided following a 60/20/20 ratio for the **Training, Validation, and Test** phases. This triple partition is essential: the validation set allows for hyperparameter optimization (using a coarse search on F1-score) and overfitting monitoring, while the test set remains completely agnostic to the learning process.

The choice of this unconventional approach is motivated by the critical need to prevent **inter-patient Data Leakage**. In a hyperspectral dataset, voxels belonging to the same cerebral volume share a strong spatial and biochemical correlation. A random split at the single-voxel level would result in signals from the same patient appearing in both the training and test sets, leading to an artificial overestimation of accuracy. By binding the atomic unit of the split to the patient, the real clinical scenario is simulated, where the system must operate on a new subject for whom it possesses no prior information. Figure 6 schematically illustrates this rotation of patients through the five validation cycles.

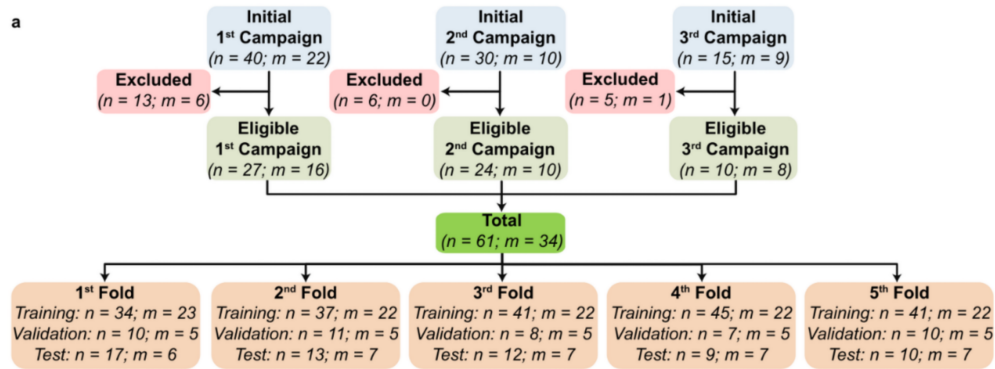


Figure 6: Diagram of the custom patient-level 5-fold cross-validation. The rotation ensures that every subject is used for testing exactly once, maintaining separation between training and validation sets for each iteration.

3.2 Training Set Optimization and Dimensionality Reduction

To reduce computational complexity and manage the high density of hyperspectral data, a subsampling strategy was adopted to reduce *training set* redundancy.

The dimensionality reduction protocol involves a first stage of **K-means Clustering** (Berthold and Hand, 2003) to identify 100 centroids per class (400 total centroids). For each centroid, the 10 most affine voxels were extracted (without replacement) using the **SAM** (*Spectral Angle Mapper*) algorithm, bringing the final dataset to a size of 4,000 voxels. The logic of SAM lies in calculating the angular distance between vectors: a zero angle implies perfect signature overlap, while an increase in the angle reflects growing spectral divergence.

In terms of performance, this operation allowed for reducing training times by about 48 times without compromising the system's predictive capability. Particularly significant is the 20% increase in accuracy detected for the **Tumor Tissue (TT)** class: this evidence suggests that selecting highly representative voxels acts as a regularizing filter, facilitating the learning of discriminative *features* even in the presence of strong initial imbalance.

3.3 Incremental Evaluation Pipeline

The adopted evaluation architecture is structured into a three-stage incremental pipeline designed to isolate and quantify the contribution of each component to the final prediction precision. This approach allows for monitoring signal evolution: from purely point-based classification to regularization based on spatial context.

The three levels of processing, described in detail in the testing phases, are structured as follows:

1. **Spectral Classification (Spectral):** Represents the system's *baseline*. The model receives the pre-processed hypercube of dimensions $H \times W \times \lambda$ as input. In this phase, one of the 8 candidate classification models is involved to generate two fundamental outputs: a *ClassMap* ($H \times W$), where each voxel is assigned to one of 4 labels, and a *probAllImage* ($H \times W \times 4$), which stores the probability distribution for each class.
2. **Spatial-Spectral Integration (Spatial/Spectral):** In this second stage, the pipeline introduces a spatial regularization component to correct any local inconsistencies. The system combines the probability map (*probAllImage*) with the first principal component (*PC1*) of the original hypercube. Processing involves applying a local **KNN-E filter**, operating on a 14×14 voxel window with $k = 40$. This filtering performs a *smoothing* of probabilities, producing maps (*knn.mapProb* and *knn.classMap*) that are significantly more spatially coherent.
3. **Refinement via Majority Voting (Majority Voting):** The final stage aims to consolidate the final segmentation by leveraging information from an unsupervised process. The pipeline integrates the map labeled by KNN with a 24-class segmentation obtained via *Hierarchical K-Means* (HKM), defined as *HKMmap*. A majority vote (*MV*) is then performed for each of the 24 clusters, stabilizing tissue boundaries and generating the system's final output, the *mv.classMap*.

The complete structure of the evaluation pipeline is shown in Figure 7.

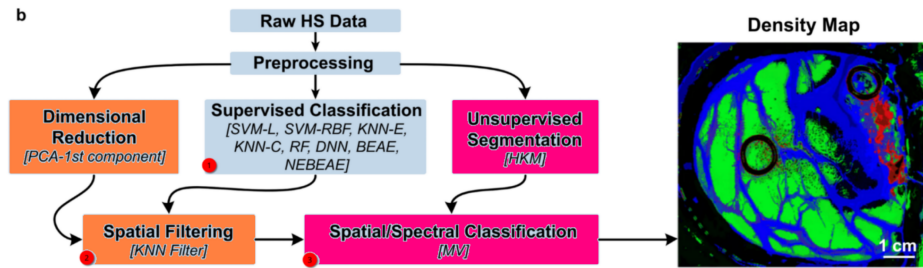


Figure 7: Block diagram of the proposed processing pipeline, integrating dimensionality reduction, supervised classification, and unsupervised segmentation.

3.4 Evaluation Metrics

To quantify the discriminative effectiveness of the models and manage the critical issues deriving from the strong class imbalance, the evaluation pipeline employs a set of metrics derived from the confusion matrix. In addition to global accuracy, it was necessary to adopt parameters capable of isolating diagnostic precision for tumor tissue.

A fundamental methodological constraint, illustrated in Figure 2, concerns the calculation of the *Macro F1-Score*: in this metric, all classes except **Background (BG)** were considered, as BG is considered irrelevant for surgical segmentation. Furthermore, to ensure the clinical significance of the tests, the evaluation was restricted exclusively to test images containing at least one voxel labeled as **Tumor (TT)**.

The metrics used are defined by the following mathematical relationships:

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \quad ; \quad MacroF1 = \frac{1}{N} \sum_{i=1}^N F1_i \quad (2)$$

$$Sens = \frac{TP}{TP + FN} \quad ; \quad Spec = \frac{TN}{TN + FP} \quad ; \quad F1 = \frac{2TP}{2TP + FP + FN} \quad (3)$$

Where TP , TN , FP , and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively. *Sensitivity* (or *Recall*) measures the system's ability to correctly identify tumor voxels, while *Specificity* evaluates precision in rejecting healthy tissue or blood vessels. Finally, the F1-score synthesizes the balance between precision and recall, providing a robust measure of segmentation quality for each single class.

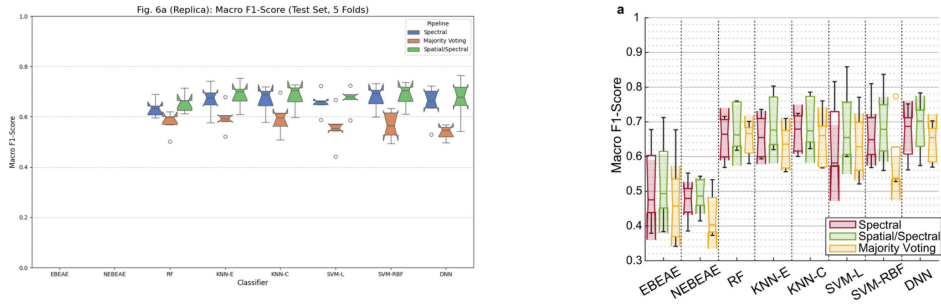


Figure 8: Comparison of Macro F1-Score distributions for different classifiers. Boxplots show performance variability across the 5 validation folds.

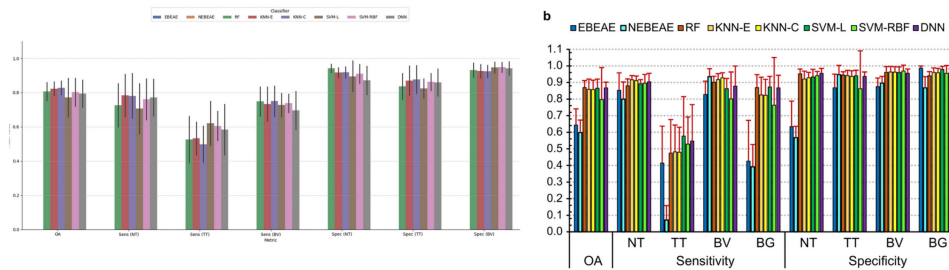


Figure 9: Detailed analysis of Overall Accuracy (OA), Sensitivity, and Specificity for different tissue classes. Note the criticality in the sensitivity of the Tumor (TT) class.

3.5 Comparative Performance Analysis

The results of the experimental replication, measured using the Macro F1-Score metric on the test set, are presented in Figure 8. The comparative analysis with the original benchmark results highlights three key points:

1. **Consistency of Trends:** Machine Learning-based models (SVM, RF) and Deep Learning (DNN) consistently outperform spectral *unmixing*-based approaches (EBEAE, NEBEAE), confirming the observations of the original study.
2. **Performance Reduction:** A systematic performance reduction (about 5%) is observed across all metrics compared to the published results. However, the variance between different folds is notably lower in our replication, suggesting a more robust and conservative estimate of generalization error.
3. **Impact of Majority Voting:** Contrary to expectations, the Majority Voting stage showed a drop in performance for the Macro F1-Score. This unexpected result might indicate critical sensitivity to unsupervised segmentation parameters or implementation specificities not fully documented in the original work.

A detailed analysis of per-class metrics (Figure 9) reveals the Achilles' heel of all examined classifiers: **Sensitivity for Tumor Tissue (TT)**. While specificity remains at excellent levels ($> 90\%$), indicating a low false positive rate, the ability to correctly identify all tumor pixels remains critical, with a high false negative rate. This limitation, persistent across all models, corroborates the hypothesis of an intrinsic difficulty in class separability at the single-pixel level, probably due to biological variability and the ground truth limitations discussed in the previous section.

4 ENSEMBLE STRATEGIES AND UNCERTAINTY ANALYSIS

Building on the limits highlighted in individual classifiers (particularly low tumor sensitivity and the inability to manage asymmetric clinical risk), this section explores the effectiveness of advanced aggregation strategies.

Three ensemble approaches were implemented and compared, each with a different operational philosophy:

1. **Probabilistic Ensemble (Soft Voting):** Aggregation based on the weighted average of probabilities, following the Sum Rule discussed in (Kittler et al., 1998), optimized to maximize global metrics.

2. **Hierarchical Risk-Sensitive Thresholding:** A rule-based approach designed to impose safety constraints on blood vessels.
3. **Clinical Stacking (Meta-Learning):** A data-driven approach based on the *Stacked Generalization* paradigm (Wolpert, 1992), using a Random Forest to learn complex correlations between model errors.

4.1 Experimental Setup

4.1.1 Weighted Soft Voting

This strategy constitutes the advanced baseline. Instead of a simple *Hard Voting* majority, which loses confidence information, the weighted average of the probability maps $P_k(x)$ of the six base models ($k = 1 \dots 6$) was calculated. Weights w_k were assigned proportionally to the *Macro F1-Score* obtained by each model during validation. The final class is determined by the maximum argument of the resulting distribution.

4.1.2 Hierarchical Thresholding (Safety-First)

To address the critical issue of intraoperative hemorrhage, a two-stage decision model reflecting the surgeon's decision-making process was developed:

- **Safety Layer (BV):** A threshold τ_{BV} is identified, calibrated to ensure Blood Vessel (BV) recall $> 98\%$. If $P(BV) > \tau_{BV}$, the pixel is classified as a Vessel, ignoring any other probability.
- **Oncological Layer (TT):** For the remaining pixels, a threshold τ_{TT} optimized to maximize the F2-Score is applied — a metric that weights tumor Recall twice as much as Precision.

4.1.3 Clinical Stacking

A *Random Forest* meta-classifier was trained, receiving as input the concatenated probability tensor of the 6 models (6x4 features per voxel). To avoid overfitting, training followed the procedures adopted in subsection 3.1 to prevent inter-patient data leakage. The model was configured with strongly asymmetric class weights (Tumor: 30, Vessels: 10, Healthy: 1) to drastically penalize false negatives in critical classes.

4.2 Quantitative Results

Figure 10 shows the performance comparison between single models and ensemble strategies.

The data analysis reveals a clear trade-off between statistical accuracy and clinical safety:

- **Soft Voting:** Confirmed as the most statistically robust strategy, reaching a **mean Macro F1 of 0.71**, higher than any single model. This method effectively averages out the noise of individual predictions, offering the best "general" segmentation.
- **Risk-Sensitive Strategies (Thresholding and Stacking):** These methods show a different behavior. Although their global F1-Score is lower (around 0.64), they excel in safety metrics. In particular, *Clinical Stacking* demonstrates a superior ability to detect infiltrating tumor (TT Sensitivity > 0.70), while *Thresholding* ensures remarkable coverage of vascular structures. The drop in F1 is attributable to a physiological increase in False Positives (reduction in specificity), which in oncological surgery could be considered an "acceptable cost" to avoid residual tumor.

4.3 Uncertainty Analysis and Quantification

To evaluate the reliability of predictions, an in-depth analysis of ensemble calibration was conducted. Unlike individual models, the *Soft Voting* architecture allows for leveraging the variance among classifiers to decompose total uncertainty into semantically distinct components, providing the surgeon with richer information on the nature of the risk.

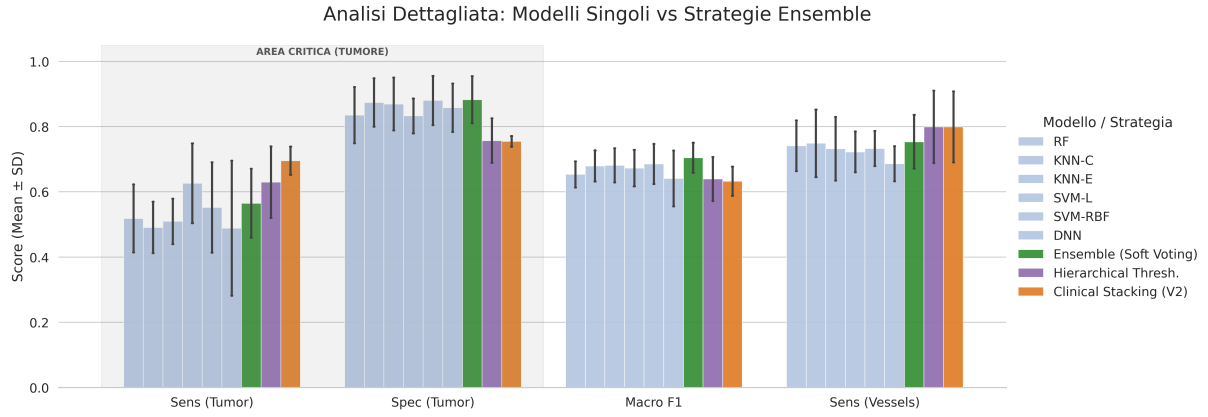


Figure 10: Final strategy comparison. Soft Voting (Green) offers the best global balance. Clinical Stacking (Orange) and Thresholding (Purple) increase tumor and vascular sensitivity, accepting a drop in specificity.

4.3.1 Uncertainty Decomposition

Let x be a voxel of the hyperspectral image and $y \in \{NT, TT, BV, BG\}$ be the target class. The ensemble consists of $M = 6$ base models (RF, SVM-L, SVM-RBF, DNN, KNN-C, KNN-E), each providing a probability distribution $P_m(y|x)$. The total predictive uncertainty was analyzed through different measures:

1. **Total Uncertainty (U_{tot}):** Represents the entropy of the ensemble's mean distribution ($\bar{P}(y|x) = \frac{1}{M} \sum P_m$). Measures how "confused" the system is overall.

$$U_{tot}(x) = H(\bar{P}(y|x)) = - \sum_{c=1}^K \bar{p}_c \log_K(\bar{p}_c) \quad (4)$$

where $K = 4$ is the number of classes and the base- K logarithm normalizes the entropy in $[0, 1]$.

2. **Aleatoric Uncertainty (U_{ale}):** Also known as Data Uncertainty, it is the mean of the individual models' entropies. It captures the noise intrinsic to the data (e.g., blurred boundaries, limited spectral resolution) that makes the sample difficult to classify for *any* model.

$$U_{ale}(x) = E_m[H(P_m(y|x))] = \frac{1}{M} \sum_{m=1}^M H(P_m(y|x)) \quad (5)$$

3. **Epistemic Uncertainty (U_{epi}):** Also known as Model Uncertainty, calculated as the difference between total and aleatoric uncertainty (equivalent to *Mutual Information* (Hüllermeier and Waegeman, 2021)). This component measures the **disagreement** among models. If U_{epi} is high, it means models are individually confident (low internal entropy) but provide divergent predictions (e.g., RF predicts Tumor, SVM predicts Vessel).

$$U_{epi}(x) = U_{tot}(x) - U_{ale}(x) \quad (6)$$

This distinction is clinically relevant: high aleatoric uncertainty suggests a physical limit of the sensor, while high epistemic uncertainty indicates an "ambiguous" zone where the models have not reached consensus, requiring maximum human attention.

4.3.2 Statistical Validation: Point-Biserial Correlation

To demonstrate that these metrics are valid risk indicators, the statistical correlation between uncertainty values (continuous) and actual classification error (binary: 0=Correct, 1=Error) was calculated. The tool used is the **Point-Biserial Correlation** coefficient (r_{pb}) (Sheskin, 2011), suitable for measuring the relationship between a continuous and a dichotomous variable:

$$r_{pb} = \frac{\mu_{err} - \mu_{corr}}{\sigma_{tot}} \sqrt{\frac{N_{err} \cdot N_{corr}}{N^2}} \quad (7)$$

Where μ_{err} and μ_{corr} are the mean uncertainties for erroneous and correct pixels, respectively, and σ_{tot} is the standard deviation of uncertainty across all pixels.

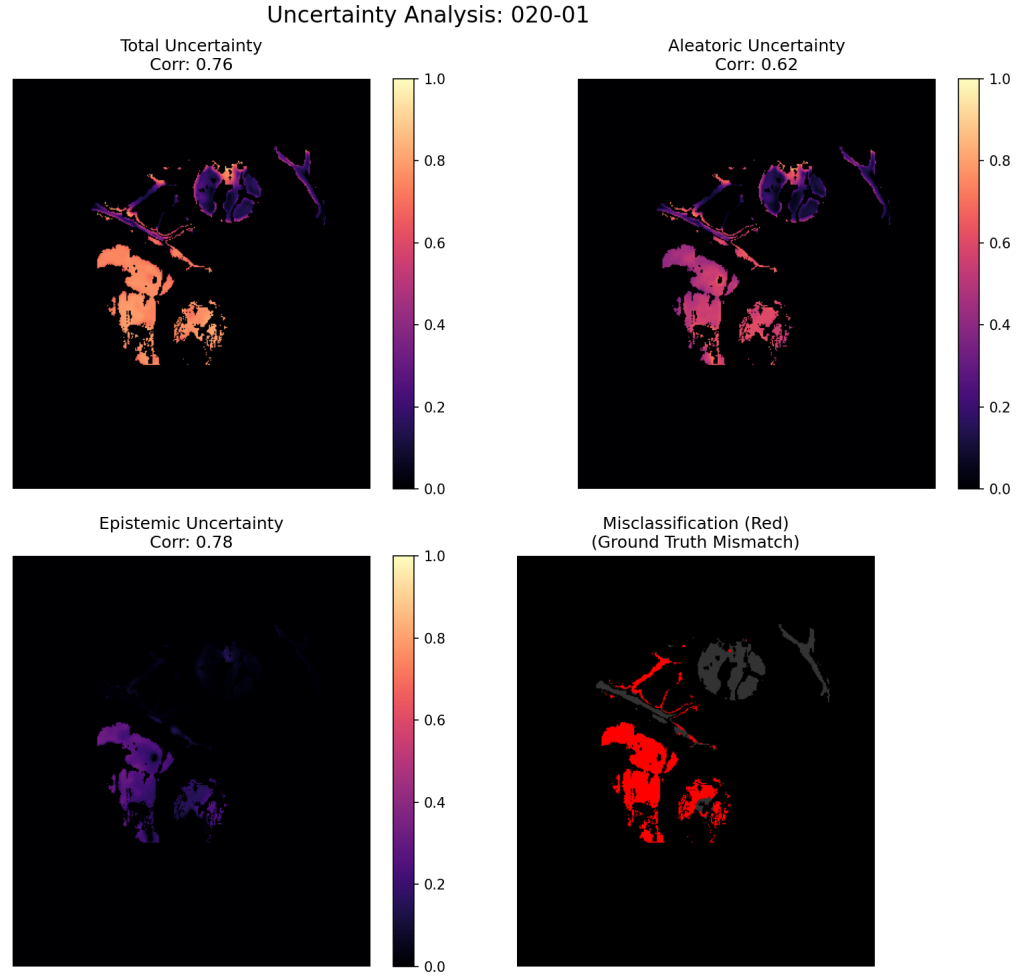


Figure 11: Detailed analysis for patient 020-01 ($r_{pb} = 0.72$). The panel shows the three uncertainty components and the error map (in red). Note how Epistemic Uncertainty (bottom left) precisely isolates the lesion edges where errors occur, signaling model disagreement.

4.3.3 Quantitative Result Analysis

The analysis was applied to the entire test dataset (Table 2). Data show a low global mean and median correlation (respectively 0.203 and 0.278 for epistemic uncertainty). Disaggregated analysis reveals significant details.

Table 2: Summary of correlation analysis (r_{pb}) for the most significant cases. Values for Total, Aleatoric, and Epistemic Uncertainty are reported, followed by global dataset statistics.

Category	Patient	Total (U_{tot})	Aleatoric (U_{ale})	Epistemic (U_{epi})
Best Cases	020-01	0.756	0.625	0.778
	034-02	0.587	0.562	0.586
	039-01	0.489	0.460	0.461
	008-02	0.431	0.360	0.546
	036-02	0.485	0.480	0.357
Worst Cases	056-02	-0.614	-0.645	-0.373
	008-01	-0.575	-0.533	-0.520
	053-01	-0.419	-0.411	-0.387
Dataset Mean		0.201	0.182	0.203
Dataset Median		0.260	0.218	0.278

In "Best Performance" cases (e.g., patient 020-01, 008-02), it is observed that **Epistemic Uncertainty** (U_{epi})

often shows a higher correlation with error than aleatoric uncertainty. This suggests that in complex scenarios, error primarily arises from model disagreement: the ensemble is "debating" over ambiguous pixels, and this debate is a stronger error predictor than simple data noise.

Conversely, "Worst Cases" (e.g., 056-02, $r_{pb} \approx -0.6$) highlight *Overconfident Error* situations: the model fails with high confidence (low entropy). A qualitative inspection of these cases might suggest the presence of systematic artifacts (e.g., specular reflections or sharp shadows) that uniformly trick all classifiers, leading to an erroneous consensus.

Figure 11 visually shows this result: uncertainty is not random noise but localizes on the tumor and blood vessel edges, offering an indispensable "risk-aware" visual guide for the operator.

4.4 Operational Validation: Risk-Rejection Curves

While point-biserial correlation analysis highlighted a heterogeneous statistical link between uncertainty and error, the system's real clinical utility emerges when evaluating performance in a *Human-in-the-loop* operational scenario.

To this end, **Risk-Rejection Curves** were generated for the three ensemble strategies (Figure 12). These graphs illustrate metric evolution as the percentage of discarded voxels varies, selected based on their Epistemic Uncertainty.

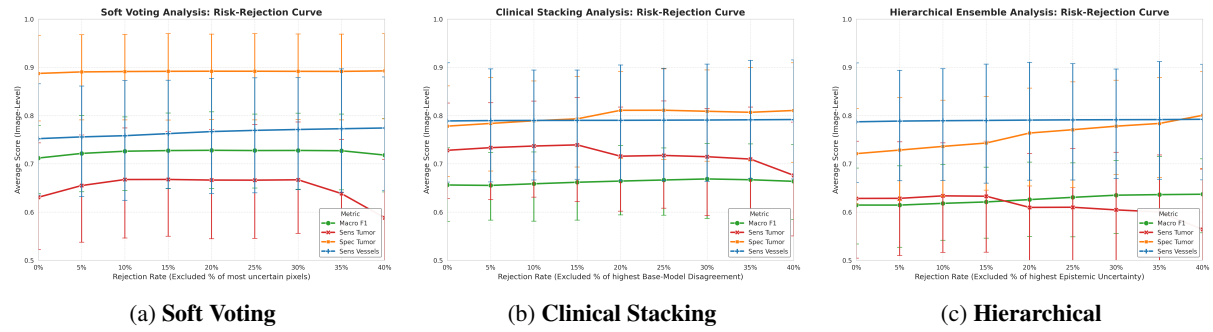


Figure 12: **Comparative Risk-Rejection Analysis.** (a) **Soft Voting** shows the greatest gain: discarding uncertain data drastically raises tumor sensitivity (red) while maintaining high specificity (orange). (b) **Clinical Stacking** starts with higher tumor sensitivity but lower specificity and shows more stable curves. (c) The **Hierarchical Ensemble** presents the lowest global performance and flat curves, indicating that rigid safety constraints limit the benefit derived from uncertainty filtering.

The comparative analysis reveals three distinct behavior profiles:

- **Soft Voting (Conservative Approach):** This strategy benefits most from uncertainty information. It starts with **excellent Specificity** (≈ 0.89 , orange line), indicating high precision in not removing healthy tissue or blood vessels, but initially lacks sensitivity. Rejection allows for recovering missed tumors (the red line rises), making it the ideal candidate for surgeries in areas where functional preservation is the priority.
- **Clinical Stacking (Aggressive Approach):** Shows opposite behavior. It starts with **markedly higher Tumor Sensitivity** (≈ 0.73 vs. 0.63 for Soft Voting) but pays for this advantage with lower specificity (≈ 0.78). The flatter curve suggests the meta-model is intrinsically robust: it has already resolved spectral ambiguities internally, making the uncertainty filter less impactful. This is the choice for maximal resection.
- **Hierarchical Ensemble (Safety Constraint):** This approach shows the lowest global metrics (Macro F1 < 0.65). Its rigid nature, based on safety thresholds for blood vessels ("Safety-First"), imposes forced decisions that epistemic uncertainty cannot effectively mitigate. The static nature of its curves confirms that the imposition of *hard-coded* rules tends to dominate over the probabilistic information provided by the ensemble.

In conclusion, epistemic uncertainty is not a universal solution but a tool that specifically empowers probabilistic architectures (Soft Voting), transforming them from moderate classifiers into highly reliable decision support systems.

5 DISCUSSION AND CONCLUSIONS

This work explored the potential of *Hyperspectral Imaging* (HSI) as an enabling technology for precision neurosurgery, addressing the critical challenge of intraoperative brain tumor segmentation. Starting from the replication of the InVivoBench benchmark, the research evolved toward the development of advanced Ensemble architectures and the integration of predictive uncertainty metrics, with the goal of overcoming the reliability limits of individual classifiers.

5.1 Summary of Results

Experimental analysis confirmed that while spectral signatures contain discriminative information, their point-based classification is hindered by significant biological variability and acquisition artifacts. In this context, the proposed aggregation strategies demonstrated clear added value:

- **Soft Voting Ensemble** proved to be the most balanced approach, ensuring high specificity and substantially benefiting from uncertainty-based filtering.
- **Clinical Stacking** offered a solution oriented toward maximizing tumor sensitivity ($\approx 73\%$), proving intrinsically robust to disagreement among base models, albeit at the cost of a higher false positive incidence.

5.2 The Uncertainty Paradox and Data Quality

A central result of this study lies in the analysis of **Epistemic Uncertainty**. Although the global statistical correlation between uncertainty and error proved low ($r_{pb} \approx 0.20$), operational analysis via *Risk-Rejection* curves revealed a more complex reality. The increase in tumor sensitivity achieved by discarding the most uncertain voxels demonstrates that the system is capable of "self-diagnosing" its shortcomings in critical areas.

However, the persistence of *Overconfident* errors and the difficulty in reaching sensitivity levels above 80% suggest a structural problem that transcends model architecture: the **Ground Truth quality**. As discussed during the dataset analysis phase, the semi-automatic labeling based on spectral similarity (SAM) might have introduced a systematic *bias*, defining sharp boundaries where a biological infiltration gradient exists. It is plausible to hypothesize that many of the "False Positives" or high-uncertainty zones detected by our ensembles are not computational errors but reflect a biochemical reality that the sparsely sampled histological "truth" fails to fully capture. The intrinsic ambiguity of some spectral signatures might therefore not be a sensor limit, but the manifestation of the diffused nature of the pathology.

5.3 Future Perspectives

In light of this evidence, future research directions should focus not only on algorithmic refinement but on redefining the validation paradigm:

1. **Ground Truth Improvement:** It is imperative to evolve toward annotation protocols that integrate dense point histology and multimodal imaging (e.g., intraoperative MRI) to validate tissue nature in spectral uncertainty zones.
2. **Human-in-the-loop Systems:** Integrating uncertainty maps into surgical displays represents the next fundamental step. Transforming the model output from a simple n-ary mask to a map exposing probabilistic risk would allow surgeons to integrate their experience where the algorithm signals ambiguity.

In conclusion, this study demonstrates that machine learning applied to HSI can become a reliable clinical tool, provided that the goal of blind automation is abandoned in favor of a *risk-aware* approach, where the machine does not just classify but engages in a dialogue with the surgeon by quantifying its own uncertainty.

References

- Berthold, M. and Hand, D. J. (2003). *Intelligent Data Analysis: An Introduction*. Springer, Berlin, Heidelberg, 2nd edition.
- Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Information Science and Statistics. Springer, New York.
- Botta, A. (2026). HSI-benchmark-eval: Codice sorgente per la segmentazione di tumori cerebrali su immagini iperspettrali. <https://github.com/AliasBotta/HSI-benchmark-eval>.
- Cristianini, N. and Shawe-Taylor, J. (2000). *An Introduction to Support Vector Machines and Other Kernel-based Learning Methods*. Cambridge University Press, Cambridge, UK.

- Fabelo, H., Ortega, S., Szolna, A., Bulters, D., Callicó, G. M., Lazcano, R., Salvador, R., Madroñal, D., Juárez, E., Villa, M., et al. (2019). In vivo hyperspectral human brain image database for brain cancer detection. *IEEE Access*, 7:39098–39116.
- Goodfellow, I., Bengio, Y., and Courville, A. (2016). *Deep Learning*. Adaptive Computation and Machine Learning. MIT Press, Cambridge, MA, USA.
- Hüllermeier, E. and Waegeman, W. (2021). Aleatoric and epistemic uncertainty in machine learning: an introduction to concepts and methods. *Machine Learning*, 110(3):457–506.
- Kittler, J., Hatef, M., Duin, R. P. W., and Matas, J. (1998). On combining classifiers. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 20(3):226–239.
- León, R., Fabelo, H., Ortega, S., Cruz-Guerrero, I. A., Campos-Delgado, D. U., Szolna, A., Piñeiro, J. F., Espino, C., O’Shanahan, A. J., Hernandez, M., et al. (2023). Hyperspectral imaging benchmark based on machine learning for intraoperative brain tumour detection. *npj Precision Oncology*, 7(1):112.
- Martín-Pérez, A., Villa, M., Rosa Olmeda, G., Sancho, J., Vazquez, G., Ortega, S., Fabelo, H., and Callicó, G. M. (2025). Slimbrain database: A multimodal image database of in vivo human brains for tumour detection. *Scientific Data*, 12(1):836.
- Sheskin, D. J. (2011). *Handbook of Parametric and Nonparametric Statistical Procedures*. Chapman and Hall/CRC, Boca Raton, 5th edition.
- Wolpert, D. H. (1992). Stacked generalization. *Neural networks*, 5(2):241–259.